3D Compositional Generative Networks for Robust Classification and 3D Pose Estimation.

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The need for OOD Testing of AI algorithms

- We typically test AI performance on IID data which comes from the same source as the training data.
- ***But it is more interesting, and insightful, to test algorithms on Out-of-Distribution (OOD) data which comes from a different source than the training data.***
- This ensures that the algorithms have deeper understanding of the data and are more likely to perform well in real world situations.
- ***Algorithms which perform well on IID data are like intelligent parrots. Feedforward deep networks are parrots, admittedly very intelligent in some cases.***

An Old Idea: Analysis by Synthesis

- Analysis by Synthesis (Ulf Grenander) proposes that to “solve” vision requires inverting the image formation process.
- ***We should study synthesis -- how images are generated– in order to perform analysis and determine what visual scene is most likely to have generated the images.***
- This can be formulated as Bayesian inference using a likelihood P(I|W) and a prior P(W).
- A modern version of this theory is inverse computer graphics.
- ***This relates to classic theories of visual perception by Helmholtz and Gregory.***

2D Compositional Generative Networks

- 2D Compositional Generative Networks (2D-CGNs).are generative models of neural feature vectors that predict the spatial patterns of neural feature activity.
- They had internal representations that corresponded (roughly) to object parts and 3D viewpoints. These representations were learnt without supervision (only object class supervision).
- 2D CGNs are effective for several tasks – object classification, occlude detection, amodal completion – and were much more robust that SOTA deep networks to occlusion and patch attack.
- But they are also limited. To go further we need models that have 3D knowledge of objects.

Humans have 3D models of objects.

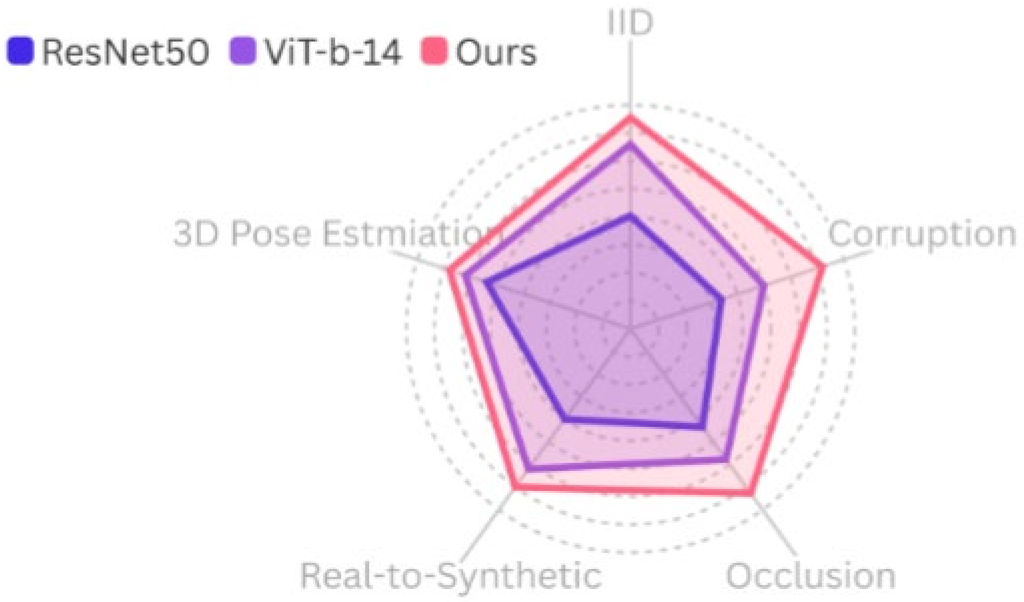
- Human vision have internal 3D representations of objects. See classic work by Shepard and Metzler (1971) on mental rotation.
- When humans recognize an object they also estimate its 3D pose and identify the positions of its parts, the boundary of the objects., etc.
- Humans probably do this in a consistent manner. I.e., they probably do not have an algorithm to recognize the object, another algorithm to estimate its rotation, another to find the parts, and so on.
- How do humans learn 3D models? As infants using multimodal cues (e.g., vision and touch) and interacting with the object.
- How can AI researchers learn 3D models of objects?

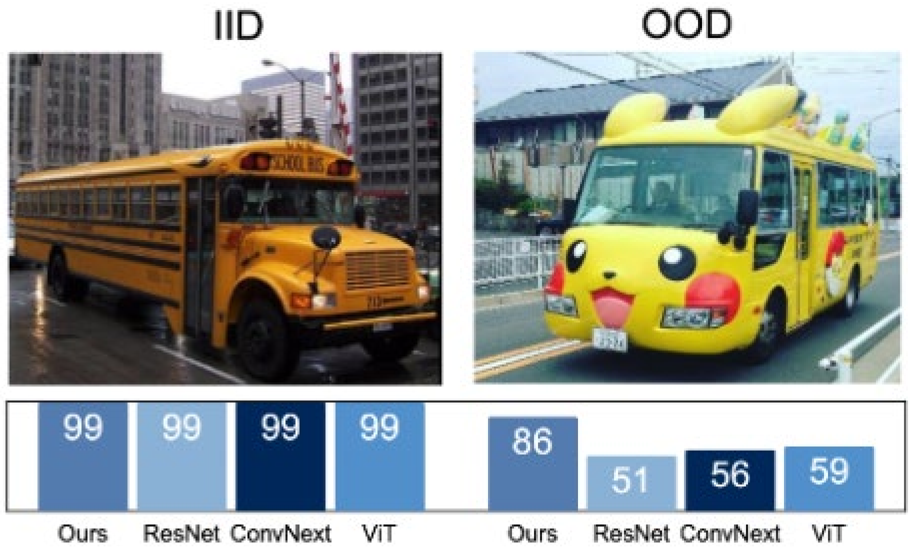
How Can AI learn 3D object models?

- There are large repositories of 3D models of different objects (ShapeNet, ObjectVerse). Objects can be represented by CAD models with meshes of vertices. (More advanced methods).
- We can learn a model for how these 3D models can generate image features (or an image).
- This requires datasets of images with 3D annotations of objects. These must specify the 3D orientation of the object and where the mesh variables project to neural features in the image,
- Three datasets: (I) Pascal 3D+, (II) ObjectNet, (III) ImageNet3D+.
- 3D-CGNs are trained and tested on Pascal3D+ (12 object classes) and on ImageNet3D+ (188 object classes).
- Can also train and test them on realistic-synthetic data generated by Computer Graphics (w. big data). These have annotations for free.

3D-CGN performance

- 3D-CGNs perform object classification and 3D-pose estimation simultaneously and are very effective when tested IID and in a variety of OOD settings. Also able to generalize to novel viewpoints.



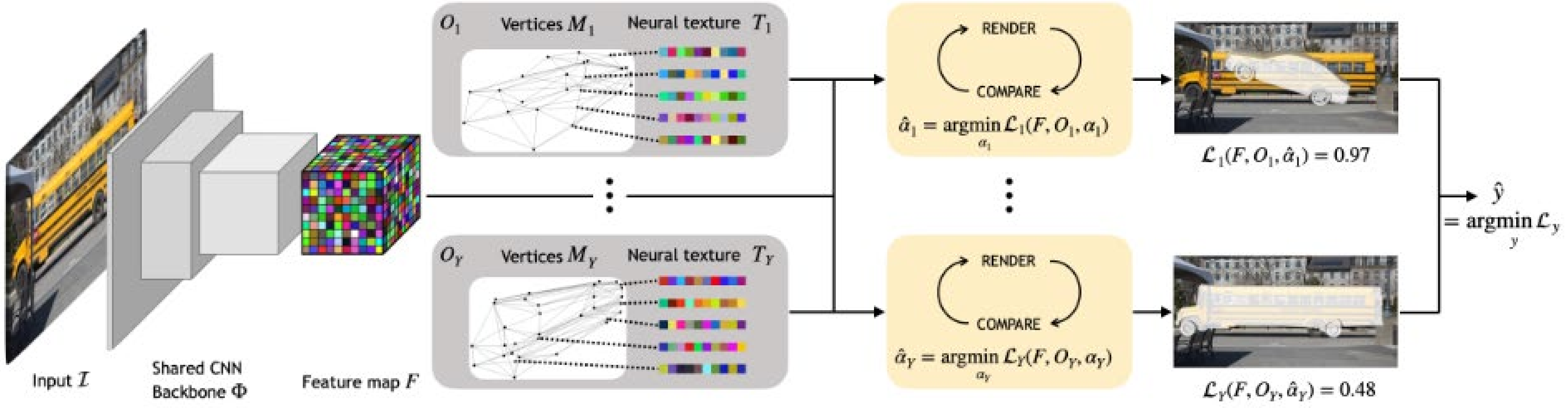


3D-CGNs series of papers

- A. Wang, A. Kortylewski, A. Yuille. ICLR. 2021. *Estimate 3D pose for single object classes robustly.*
- A. Jesslen et al. EECV. 2024. *Classifies Objects and Estimates 3D pose robustly for 12 object classes (Pascal3D+)*
- X. Yuan et al. In review. 2024. *Classifies Objects and Estimates 3D pose robustly for 188 object classes (ImageNet 3D+).*
- P. Kaushik et al. 2024. UDA transfer. Improves the ability of 3D-CGNs to generalize from the training set. (E.g., from real to synthetic).

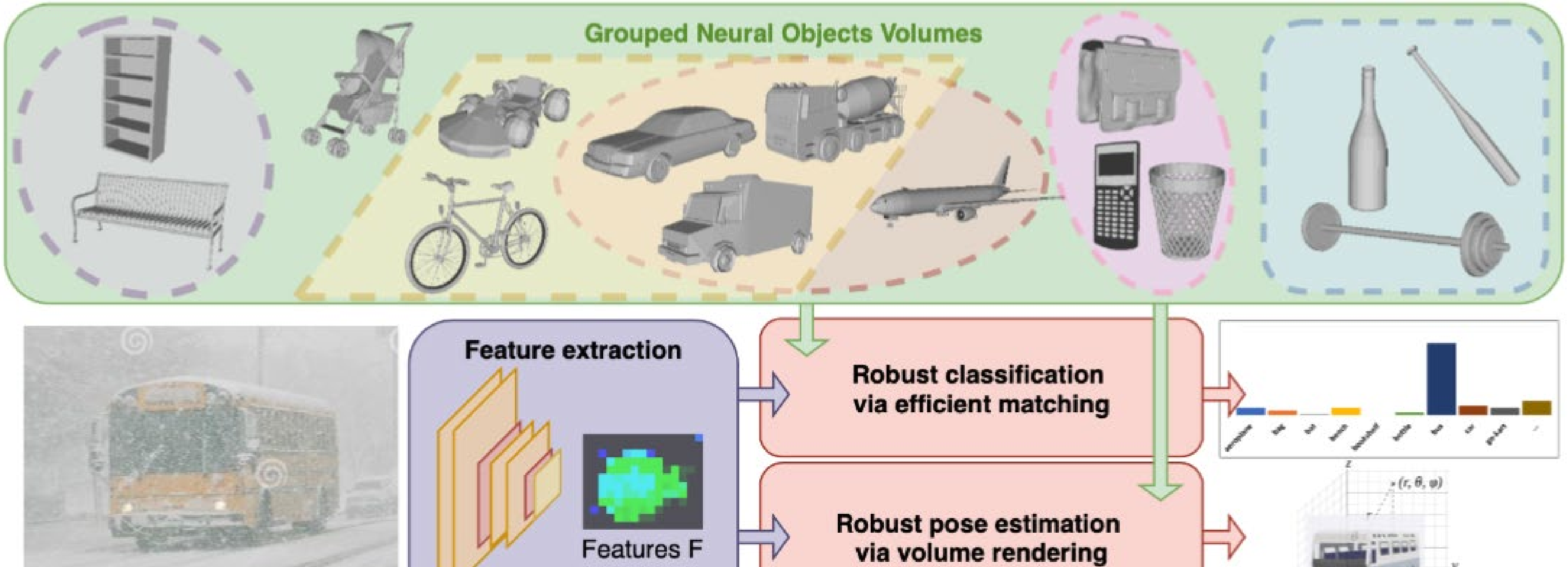
Overview of 3D-CGNs

- For each image, the network extracts a feature map F . Note that for illustrative purposes, we visualize a projection of a CAD model but we use cuboid meshes with much lower detailed geometry.
- Render and compare.



Scaling up to 188 classes.

- The top-green box represents the variety of objects (188)that we consider and illustrates the grouping of Neural Vertex Features (NVF) (with arbitrary groups for illustrative purposes). The lower part illustrates the inference pipeline. During inference, an image is first processed into a feature map F (purple box). Matching F to NVF, predicts object class (top-red box), and 3D pose (bottom-red box)



Learning the Model (1):

- We learn Neural Features for each object vertex.
- We use supervised contrastive learning which is trained to distinguish between a vertex and other vertices on the object (outside a neighborhood) and vertices on other objects.
- Here the distribution $P(f_{k\to i}|C_k) = c_M(\kappa)e^{\kappa f_{k\to i}\cdot C_k}$, with C_k as the mean
- This is Fisher von-Mises. The mean is learnt from each vertex and the network trained by maximizing:

$$\frac{P(f_{k\to i}|C_k)}{\sum_{\substack{C_l\in\mathcal{C}_y\\ C_l\notin\mathcal{N}_k}} P(f_{k\to i}|C_l) + \omega_\beta \sum_{\beta_n\in\mathcal{B}} P(f_{k\to i}|\beta_n) + \Omega_{y,\bar{y}} \quad \omega_{\bar{y}} \sum_{\substack{C_m\in\mathcal{C}_{\bar{y}}\\ \bar{C}_m\sim U(\bar{C}_m)}} P(f_{k\to i}|\bar{C}_m)},$$

Learning the Model (2).

- Training the model requires knowing where each vertex is projected for each image (provided by Pascal3D+, ImageNet3d+).
- Each vertex has a learned feature vector – its mean.
- We also learn the backbone (CNN or Transformer) which outputs the feature vectors. This training encourages the feature vectors of each vertex to be invariant to viewpoint and object instance.
- Note: we train the model to make the feature vectors (means) at each vertex different from each other. We do not train for the tasks which we evaluate (classification or 3D pose).
- Caveat: there are more details including how to ensure that this training is efficient and scales with number of object classes.

Inference (1)

- We now have a probability model for generating the feature vectors on an object conditioned on the object class and the 3D pose (by contrast 2D-CGNs used a mixture to deal with pose).
- We introduce a z-variable to deal with occlusion.
- The probability model is

$$\prod_{f_i \in F} P(f_i | z_{i,k}, y) = \prod_{f_i \in F} P(f_i | C_k)^{z_{i,k}} \prod_{f_i \in F} \max_{\beta_n \in \mathcal{B}} P(f_i | \beta_n)^{1 - z_{i,k}}.$$

- To perform inference we must search over object class and viewpoint to find the object that best predicts the input image.

Inference (2)

- **Render and Compare: (3D pose)**
- Basic Idea: make initial guess of object and 3D pose. Predict the input image, compare to image, adjust 3D pose and object class.
- In first paper (A. Wang et al. 2021) we developed an efficient steepest descent strategy which estimated the 3D pose (if object class known), Technically complex (train features to avoid local minima)
- **Proposal Method: (Object Classification)**
- Use a simpler model which is fast to compute to compute. E.g., classify object based on detected feature vectors.

$$S_y = \sum_{f_i \in F} \max\{\max_{C_k \in \mathcal{C}_y} f_i \cdot C_k, \max_{\beta_n \in \mathcal{B}} f_i \cdot \beta_n\}.$$

Results of classification on ImageNet3D+

- Results: (Some not updated).
- IID. Synthetic. Occlusion. Corruption

Models	IID	Synthetic
Resnet50	84.8	58.2
ViT-b-14	92.4	72.3
NOVUM	85.7	68.8
OURS (ResNet50)	86.5	69.3
OURS (ViT-b-14)	**93.5**	**74.9**

Table 1: Classification Results on Imagenet3D+

Classification under Occlusion and Corruption

Model	L0	L1	L2	L3	**Average**	brightness	frost	snow	fog	**Average**
Resnet50	84.8	58.8	34.7	11.2	33.9	71.4	37.5	19.2	63.9	48.0
ViT-b-14	86.2	60.1	35.9	12.9	36.3	68.3	48.3	21.8	64.3	50.7
NOVUM	85.7	64.6	37.6	13.4	38.5	75.1	46.1	30.1	72.6	55.9
OURS	**88.2**	**65.2**	**37.8**	**13.5**	**38.8**	**78.8**	**48.6**	**30.4**	**73.6**	**57.9**

able 3: Classification results on clean, occluded, and corrupted ImageNet3D+. Different occlusi
evel (L1, L2, L3) and different corruption type applied.

Results of 3D pose on ImageNet3D+

Pose Estimation under Occlusion and Corruption

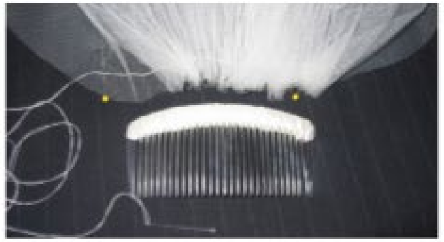
Model	L0	L1	L2	L3	brightness	snow	frost	fog
Resnet50	55.6	40.4	27.5	14.4	50.8	29.1	38.7	51.3
ViT-b-14	56.9	42.7	28.2	15.7	52.2	30.4	40.9	51.6
NOVUM	57.2	42.6	28.8	15.6	51.9	32.5	41.0	52.7
OURS	**57.6**	**43.4**	**29.2**	**16.0**	**54.1**	**32.7**	**42.2**	**53.9**

Note 10 object classes have ambiguous 3D pose.

Better pose estimates if they are removed.

3D Pose Estimation

	IID	Occ.	Corr.
All classes	57.6	29.5	45.7
w/o Elongated	**59.3**	**32.8**	**48.3**



comb

fork

pen

pencil

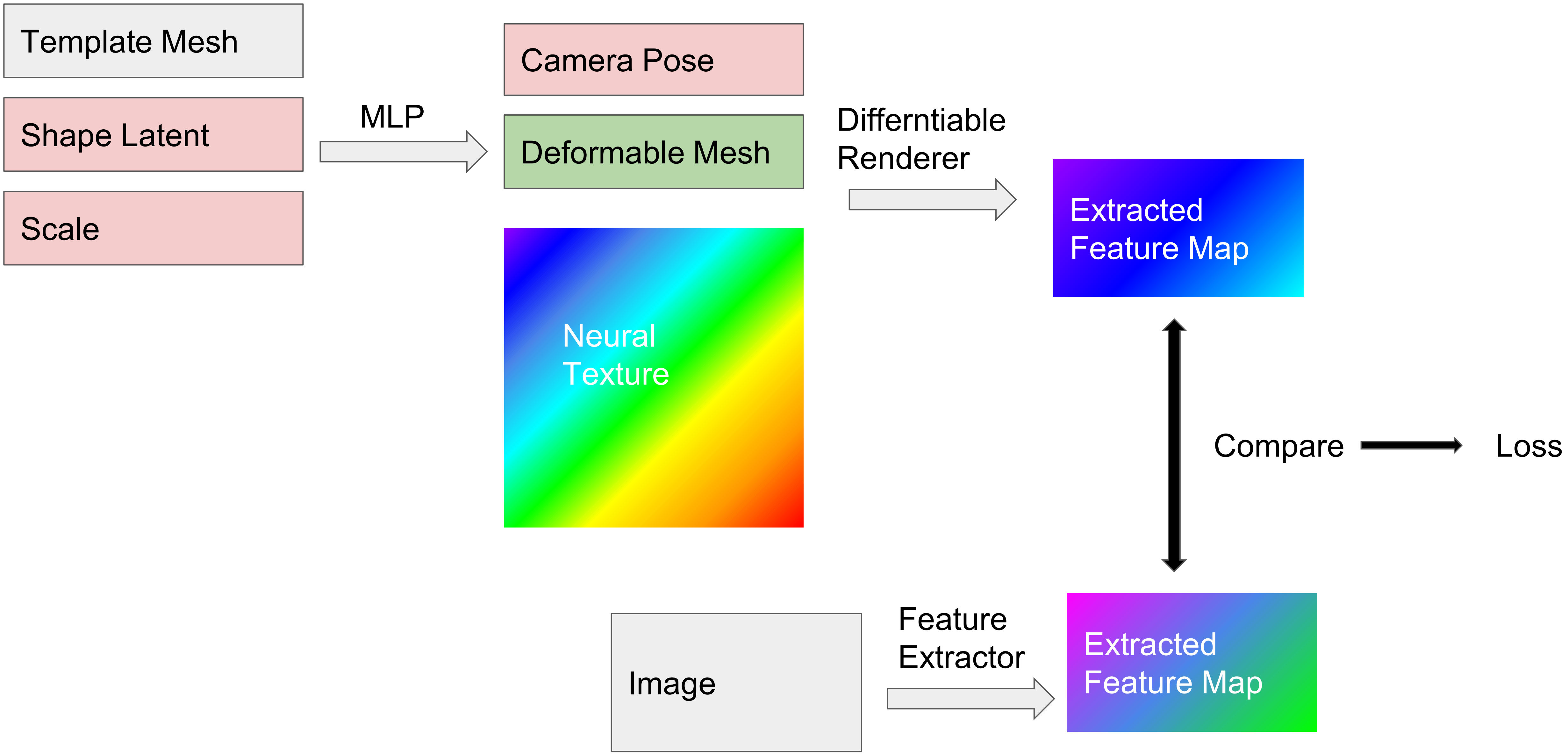
Summary

- 3D-CGNs are very effective and robust for object classification and 3D pose estimation compared to standard deep networks,
- Their ability to transfer OOD can be improved by exploiting their compositional structure (P. Kaushik et al. 2024).
- This is evidence that object models with explicit 3D representations are more effective than models which do not.
- Caveat: 3D-CGNs require datasets with 3D annotations. These are limited – we had to create ImageNet3D+. Synthetic datasets with 3D groundtruth are easier to create.
- The 3D-CGNs only use crude 3D representations (cuboid) and features. They are only the starting point.

Extra Material

- The remaining slides give more details and results about 3D CGNs

Inference by Render and Compare: (recent alternatives)



3D Generative Models for classification and 3D pose

occ Level	Clean	L1	L2	L3
Resnet50-general	**99.30**	93.81	77.78	45.21
NeMo	88.04	72.51	49.31	22.27
Ours	98.79	**95.81**	**85.63**	**57.91**

Table 1. P3D+ Classification Result

Metric	$ACC_{\frac{\pi}{6}} \uparrow$				$ACC_{\frac{\pi}{18}} \uparrow$			
occ Level	L0	L1	L2	L3	L0	L1	L2	L3
Res50	82.4	65.6	45.1	21.4	38.8	24.7	14.5	5.1
NeMo	82.4	62.1	39.1	13.3	60.1	38.9	21.1	5.2
Ours	**85.1**	**72.9**	**53.1**	**29.9**	**60.4**	**41.5**	**24.8**	**10.5**

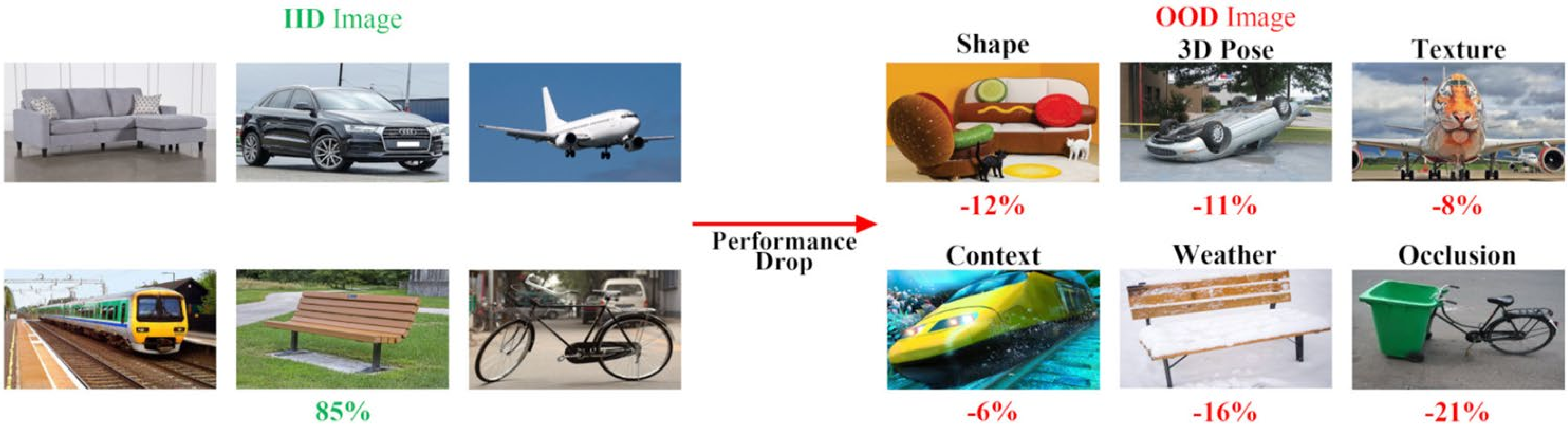
Table 3. P3D+ 3D-Classification

L1: 20-40%
L2: 40-60%
L3: 60-80%

- **Normal classification**: 3D generative models achieve **similar results** to deep neural networks
- **Occlusion classification:** 3D generative models perform much **better** than deep neural networks

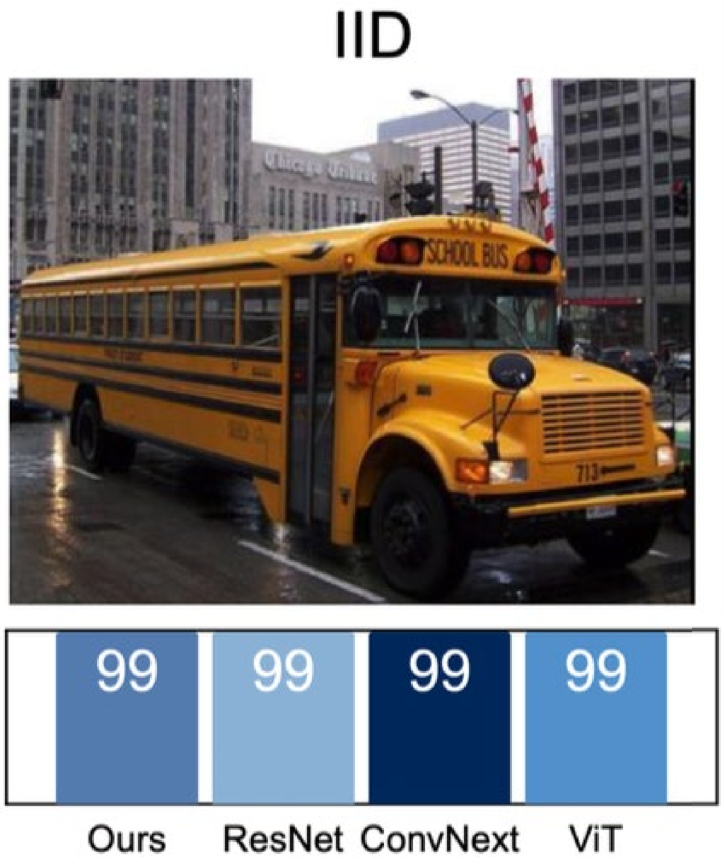
Reference
[1] Zhang, Guofeng, et al. “Inverse Rendering of Discriminative Neural Textures of 3D-aware Image Classification.” In review 2023.

Testing on IID data vs. OOD data: standard deep networks have a big drop in performance



Reference
[1] Zhao, Bingchen, et al. "OOD-CV: A Benchmark for Robustness to Individual Nuisances in Real-World Out-of-Distribution Shifts." ICML 2022 Shift Happens Workshop. 2022.

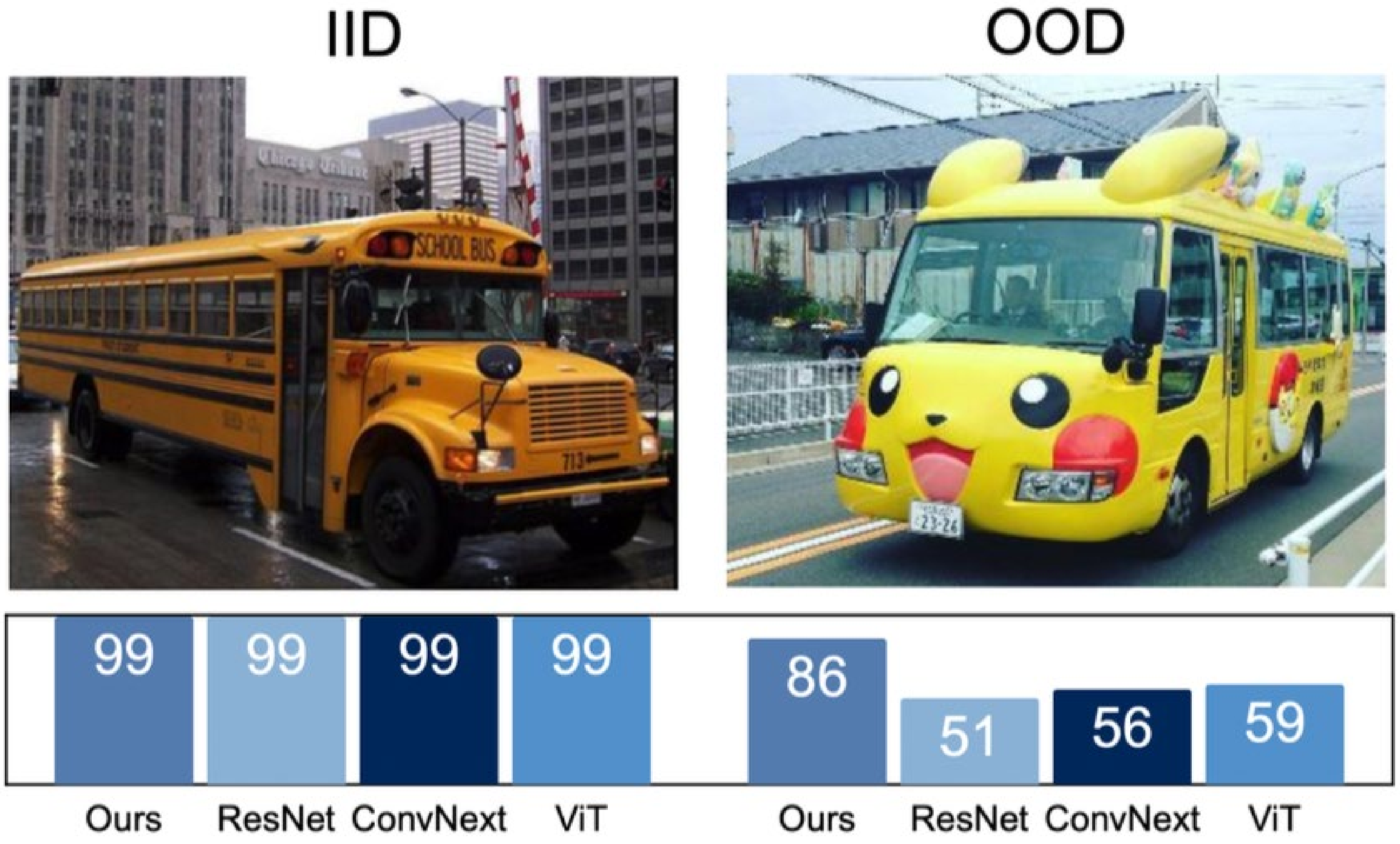
Background: deep networks work well on IID data



Results show object classification accuracy obtained on the OOD-CV dataset [1].

Reference
[1] Zhao, Bingchen, et al. "OOD-CV: A Benchmark for Robustness to Individual Nuisances in Real-World Out-of-Distribution Shifts." ICML 2022 Shift Happens Workshop. 2022.

Deep networks don’t work well on OOD data



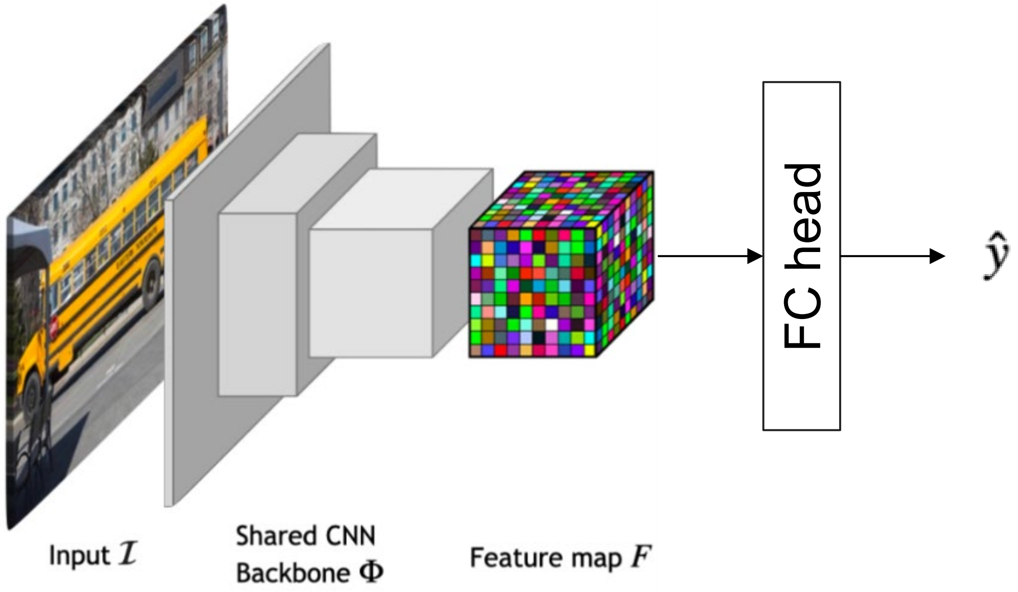
Results show object classification accuracy obtained on the OOD-CV dataset [1].

Reference

[1] Zhao, Bingchen, et al. "OOD-CV: A Benchmark for Robustness to Individual Nuisances in Real-World Out-of-Distribution Shifts." ICML 2022 Shift Happens Workshop. 2022.

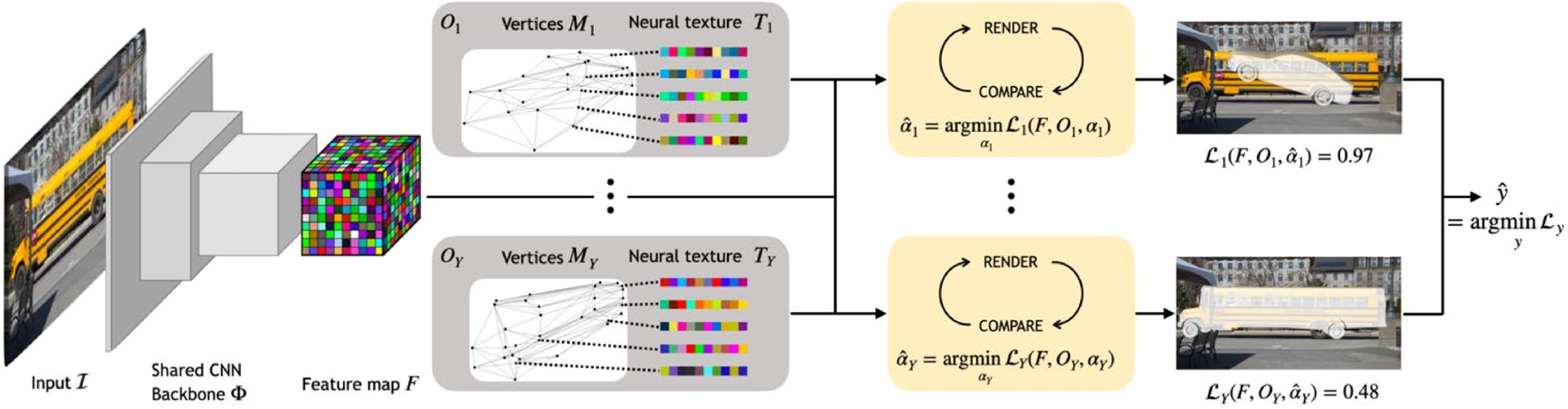
Why deep networks don't work well on OOD data?

Reason: traditional deep networks do not use **the 3D structure of objects**



3D Generative models perform better deep networks

3D Generative models exploit the **3D structure** of the object



Classification results:
3D Generative Models have SOTA performance on OOD and occluded data

Dataset	P3D+	occluded-P3D+				OOD-CV					
Nuisance	L0	L1	L2	L3	Mean	Context	Pose	Shape	Texture	Weather	Mean
Resnet50	99.3	93.8	77.8	45.2	79.6	45.1	61.2	55.2	48.3	47.3	51.4
Swin-T	**99.4**	93.6	77.5	46.2	79.7	63.0	71.4	65.9	61.4	59.6	64.2
Convnext	**99.4**	95.3	81.3	50.9	81.8	53.6	61.2	60.8	57.2	47.1	56.0
ViT-b-16	99.3	94.7	80.3	49.4	80.9	57.8	67.3	61.0	54.7	54.5	59.0
Ours	99.1	96.1	86.8	59.1	85.3	**85.2**	**88.2**	**84.6**	**90.3**	82.4	**86.0**
Ours++	**99.4**	**96.8**	**87.2**	**59.2**	**85.7**	85.1	88.1	84.1	88.5	**82.7**	85.4

Classification results:
3D Generative Models have high performance on corrupted data

Dataset	P3D+	corrupted-P3D+												
Nuisance	L0	defocus blur	glass blur	motion blur	zoom blur	snow	frost	fog	brightness	contrast	elastic transform.	pixelate	jpeg	mean
Resnet50	99.3	67.6	41.4	73.5	87.5	84.4	84.3	93.9	98.0	90.0	46.4	82.1	95.5	78.7
Swin-T	**99.4**	60.7	37.1	70.9	81.3	88.5	91.6	95.4	97.9	92.1	56.3	79.2	95.3	78.9
Convnext	**99.4**	70.1	58.7	76.5	**90.0**	**92.3**	92.9	**98.5**	**99.2**	**98.4**	67.6	84.2	**98.7**	85.6
ViT-b-16	99.3	64.5	**78.1**	80.3	88.2	91.2	**94.1**	90.5	98.7	85.1	84.8	**96.9**	**98.7**	87.6
Ours	99.1	**90.1**	66.9	86.8	84.9	81.3	88.1	98.2	97.9	96.8	**96.7**	**96.9**	98.1	90.2
Ours++	**99.4**	89.9	66.4	**87.3**	87.2	83.3	89.8	98.4	98.0	96.9	96.5	96.7	98.4	**90.4**

Pose estimation results:
3D Generative Models are SOTA on OOD-CV

	Dataset	P3D+	occluded-P3D+				OOD-CV					
	Nuisance	L0	L1	L2	L3	Mean	Context	Pose	Shape	Texture	Weather	Mean
	Resnet50	33.8	22.4	15.8	9.1	15.8	15.5	12.6	15.7	22.3	23.4	18.0
	Swin-T	29.7	23.3	15.6	10.8	16.6	18.3	14.4	16.9	21.1	26.3	19.8
	Convnext	38.9	22.8	12.8	6.6	14.1	18.1	**14.5**	16.5	21.7	26.6	19.9
$ACC_{\frac{\pi}{18}} \uparrow$	ViT-b-16	38.0	23.9	13.7	7.4	15.0	24.7	13.8	15.6	25.0	28.3	21.5
	NeMo	62.9	**45.0**	**30.7**	**14.6**	**30.1**	21.9	6.9	19.5	34.0	30.4	21.9
	Ours	61.6	42.8	27.0	11.6	27.2	**23.6**	10.4	**22.7**	37.5	**35.5**	**25.5**
	Ours++	**65.1**	**45.0**	28.7	12.5	28.8	23.5	9.8	22.3	**37.9**	34.5	24.8
	Resnet50	82.2	66.1	53.1	42.1	53.8	**57.8**	34.5	50.5	**61.5**	60.0	51.8
	Swin-T	81.4	58.5	47.3	38.8	48.2	52.3	41.1	45.7	50.1	64.9	50.9
	Convnext	82.4	63.7	47.9	36.4	49.3	51.7	**43.4**	44.8	48.0	**65.9**	50.7
$ACC_{\frac{\pi}{6}} \uparrow$	ViT-b-16	82.0	65.4	49.5	37.6	50.8	54.7	34.0	49.5	59.1	59.0	51.3
	NeMo	87.4	75.9	**63.9**	**45.6**	**61.8**	50.3	35.3	49.6	57.5	52.2	48.0
	Ours	86.1	74.8	59.2	37.3	57.1	54.3	38.0	**53.5**	60.5	57.3	**51.9**
	Ours++	**89.8**	**78.1**	62.6	39.0	59.9	55.4	36.7	53.1	60.0	57.0	51.4

Table S3: Pose Estimation results on (occluded)-PASCAL3D+, and OOD-CV dataset. Pose accuracy is evaluated for error under two thresholds: $\frac{\pi}{6}$ and $\frac{\pi}{18}$ separately. Noticeably, RCNet has equivalent performances to current SOTA for 3D-pose estimation event hough it has not been specifically designed for this task.

Qualitative results on Occluded PASCAL3D+ and OOD-CV

(a) L2 Occluded, Car

(b) Context OOD, Bicycle

(c) Weather OOD, Aeroplane

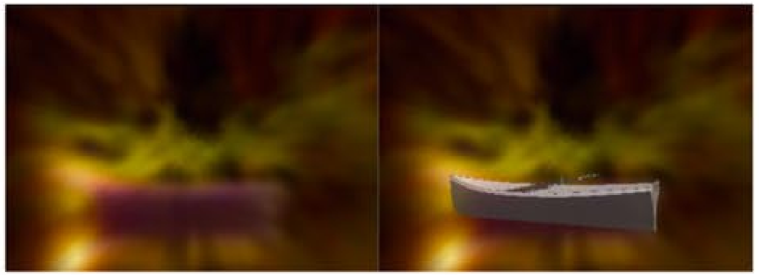
(d) Texture OOD, Sofa

(e) Shape OOD, Bus

(f) Pose OOD, Motorbike

3D Generative Models are **robust** to **occluded** data and **OOD** data

Qualitative results of on Corrupted PASCAL3D+



(g) Zoom Blur Corruption, Boat

(h) Motion Blur Corruption, Table

(i) Frost Corruption, Tvmonitor

(j) Snow Corruption, Chair

(k) Brightness Corruption, Bottle

(l) Fog Corruption, Bus

Generative 3D models are **robust** to **corrupted data**

Conclusion

- It is important to test AI on OOD data. Algorithms that perform well on IID data can degrade badly on OOD.

- Approximate Analysis by Synthesis is an old idea which has a lot of potential for computer vision and has been proposed as a theory for human visual perception.

- 3D generative models and occluder processes enables robustness to occluded data.

- They can also generalize to out-of-distribution (OOD) data needing minimal modification.

- This is only the starting point. 3D generative models work only on a limited number of objects. There are many technical ways to improve them.

- *Work with A. Kortylewski, A. Jesslen, A. Wang, W. Ma, G. Zhang and many others.*